

SmartSenseCNC

Real-Time CNC Machine Downtime Detection & Predictive Analytics using MYOSA IoT

MYOSA IoT Innovation Contest | Team Tech Titans | MMIT, Pune

1. Expression of Interest

In a typical small-to-medium CNC workshop, **2–3 hours per shift** are lost to undetected machine idle time — translating to an estimated **■25,000–40,000/month** in wasted operating costs per machine. Despite being a well-understood problem, most SME workshops still rely on manual observation, leaving downtime largely invisible.

Team Tech Titans proposes **SmartSenseCNC**: a low-cost, data-driven IoT solution that continuously monitors CNC machine states (OFF / IDLE / ACTIVE) using the MYOSA Mini IoT Kit augmented with an SCT-013 current clamp sensor, a rule-based state machine, and a real-time web dashboard. By fusing vibration, current, and light data against calibrated thresholds, the system accurately detects downtime in real time and surfaces actionable insights — helping workshop owners make evidence-based decisions to recover lost productivity.

Team Members

Name	Year	Institute	Email	Contact
Gargi Shinde	Third Year	MMIT, Pune	gargishinde@ieee.org	9370746839
Soham Shinde	Third Year	MMIT, Pune	soham.d.shinde@ieee.org	8668475432
Vedika Mudhliar	Third Year	MMIT, Pune	vedikamudhliar@ieee.org	7666962576
Faculty Mentor	Institute	Email	Contact	
Dr. Monika Dangore	MMIT, Pune	monika.dangore@ieee.org	9545554530	

2. System Architecture & MYOSA Platform Usage

SENSING LAYER
MPU6050 Accelerometer APDS9960 Light Sensor SCT-013 Current Sensor
▼ Wired I2C / Analog
PROCESSING LAYER — MYOSA Board (ESP32)
Rule-Based State Machine (OFF / IDLE / ACTIVE) → Raw Sensor Streaming via MQTT → ML Classifier on Flask Backend
▼ Wi-Fi (MQTT / HTTP)
DASHBOARD & ANALYTICS LAYER
Flask Web App Live State Display Downtime Trend Charts CSV Export
▼ On-site Feedback
ALERT LAYER
OLED Display SSD1306 (Live Status) Threshold-based Buzzer Alert

Component Roles

Component	Role in System	Why it matters
MPU6050 (MYOSA)	Detects spindle vibration patterns	Primary signal — high vibration = ACTIVE machining

APDS9960 (MYOSA)	Reads machine indicator light colour	Confirms IDLE/ERROR states via green/red light
SCT-013 (External)	Non-invasive current clamp on CNC power line	Most reliable ACTIVE vs IDLE discriminator — no wiring modification needed; safe clip-on installation
ESP32 (MYOSA)	Streams raw sensor data via Wi-Fi (MQTT)	Sends readings to Flask backend every 5s for ML classification
OLED SSD1306 (MYOSA)	Local live status display	Shows current machine state (ACTIVE/IDLE/OFF) on-site without needing dashboard
Flask Dashboard	Visualises live status + historical trends	Makes data actionable for workshop owners without technical background

3. Additional Software, ML Classifier & Data Collection

Data Collection — Real CNC Machine

Sensor readings will be logged from an actual CNC machine at our institute's workshop across three labelled operational phases — OFF, IDLE (spindle on, no cut), and ACTIVE (cutting in progress). Each reading captures accelerometer XYZ magnitude, AC current RMS from the SCT-013 (via ESP32 ADC), and light sensor intensity at 10 Hz. A minimum of 500 labelled samples per class will be collected to train and validate the classifier.

ML Classifier — Random Forest (on Flask Backend)

A Random Forest classifier (scikit-learn) will be trained on the collected CNC data to automatically detect machine states without manually setting thresholds. The model runs on the Flask backend — the ESP32 streams raw sensor readings via Wi-Fi, and the backend classifies the state in real time and updates the dashboard.

Aspect	Detail
Model	Random Forest (scikit-learn)
Features	Vibration RMS, Current RMS, Light Level
Output	State label — OFF / IDLE / ACTIVE, updated every 5 seconds
Validation	80/20 train-test split; target accuracy $\geq 92\%$ on real CNC data
Deployment	Flask backend — no embedded constraints, easy to retrain per machine

Software Stack

Layer	Technology	Deliverable
Firmware	Arduino C++ on ESP32 (MYOSA)	Raw sensor streaming via MQTT every 5 s
ML Model	Python scikit-learn (Random Forest)	State classification on Flask backend
Backend	Python Flask + SQLite	REST API, data storage, alert logic
Frontend	HTML/CSS + Chart.js	Live state panel + daily downtime bar chart
Data Export	CSV download endpoint	Shift-wise downtime reports for workshop owner

4. Significance, Innovation & Impact

- **Quantified problem:** A single idle CNC machine wastes ■25,000–40,000/month in SME workshops. Our system targets a 35–40% reduction in undetected idle time through real-time alerting.
- **Three-sensor fusion with ML:** Combining accelerometer + light sensor + SCT-013 current clamp as features for a Random Forest classifier gives far higher accuracy than any single

sensor or manual threshold — and adapts across different machines without recalibration.

- **Non-invasive installation:** The SCT-013 clamp attaches to the power cable without any wiring modification, making the system safe and easy to deploy on existing machines.
- **ML on Flask backend:** Running inference server-side avoids embedded deployment complexity while still delivering real-time state updates every 5 seconds to the dashboard.
- **Accessible analytics:** The Flask dashboard presents downtime trends in plain language, making smart manufacturing genuinely usable for non-technical SME operators.

5. Conference Demo Plan

The live demo will showcase the complete end-to-end pipeline. The MYOSA board connected to MPU6050, APDS9960, and SCT-013 will be mounted on a benchtop motor rig that reproduces the three CNC states. Attendees will see: (1) real-time state transitions on the OLED (SSD1306) display as the motor is switched between states; (2) the Flask dashboard on a laptop showing live status and a pre-recorded 8-hour downtime trend from actual CNC machine data; (3) a threshold-breach alert (buzzer + dashboard notification) triggered when IDLE time exceeds 5 minutes. A printed A3 poster will illustrate the system architecture and key results — including before/after downtime comparisons from the real machine trial.

SmartSenseCNC demonstrates that intelligent, affordable IoT monitoring is achievable today — without expensive industrial systems — directly empowering India's SME manufacturing sector under the Industry 4.0 vision.